

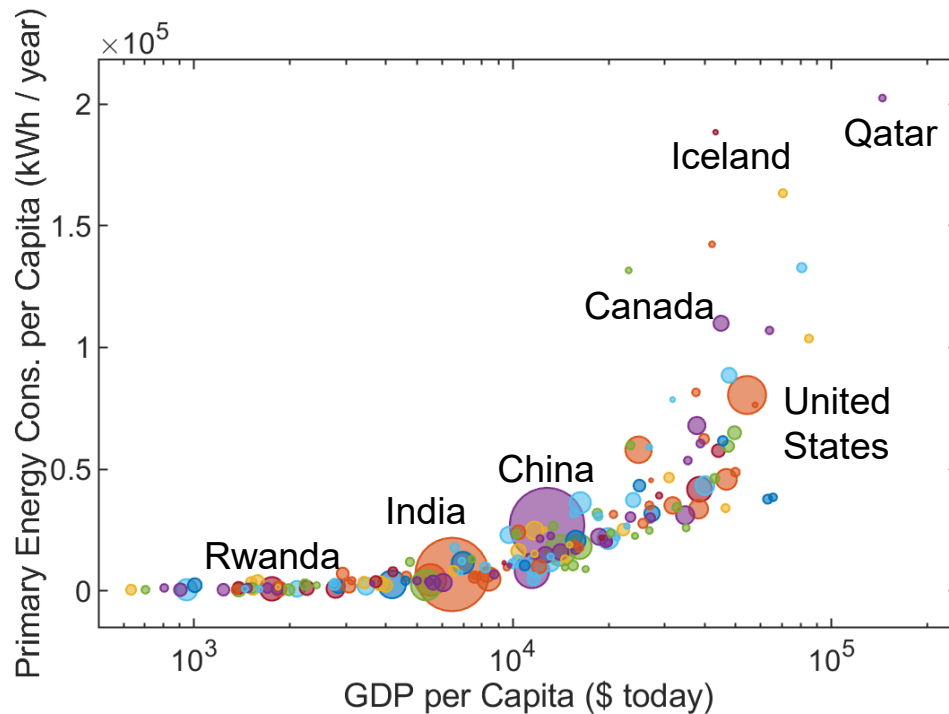
Thermodynamics

Lecture 1B:

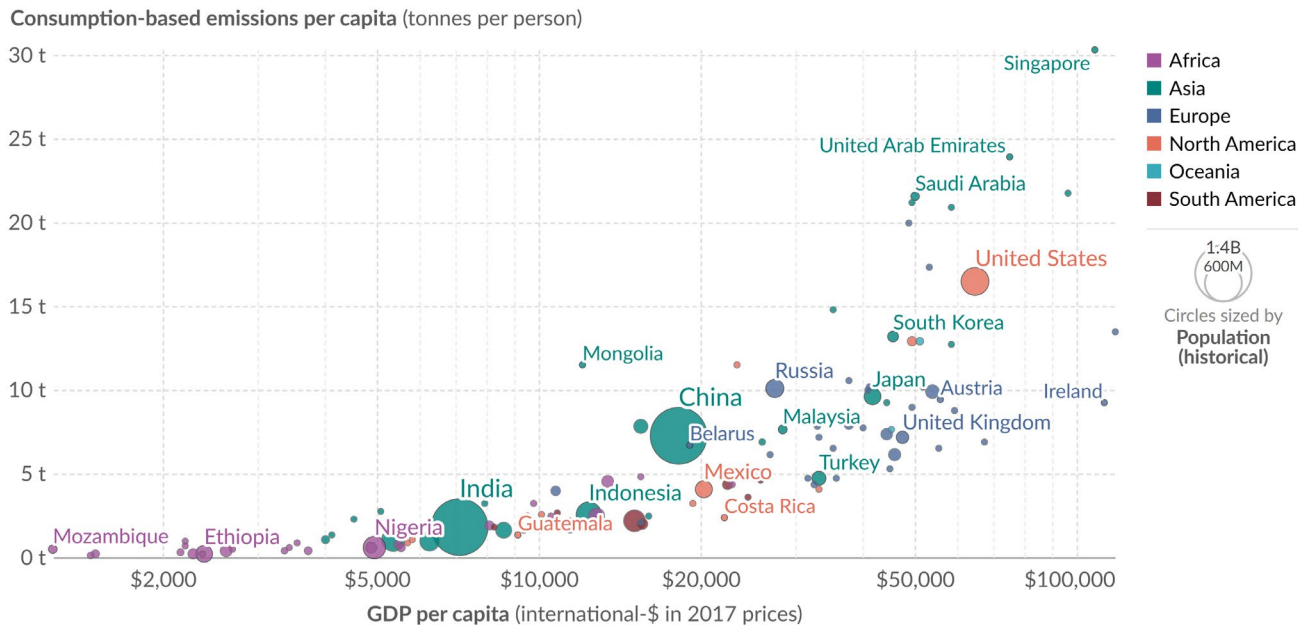
- Motivation
- Basic Concepts
- System properties

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Global energy demand



Energy and CO₂ emissions



Data source: Global Carbon Budget (2024); Population based on various sources (2024); World Bank (2023)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

Thermodynamics: relevance



Airplanes



Power plants



Cars

Thermodynamics: relevance



Food processing



Refrigeration



Heating

Thermodynamics challenges in the future

- Electrification
- Decarbonization
- Artificial Intelligence
- Emerging technologies:
 - Atmospheric water harvesting
 - Passive radiative cooling
 - ...

Learning objectives

What you will be able to do after this course:

- I. *Knowledge:* Write the first and second laws of thermodynamics.**
- II. Comprehension:* Describe the first and second laws in the student's own language.
- III. Application:* Solve simple single-answer problems using the first law.
- IV. Analysis:* Solve problems requiring both the first and second laws.
- V. Synthesis:* Develop schemes to improve the efficiency or performance of thermodynamic systems, e.g., compressors, turbines, heat pumps, etc.

SI Units

Dimension	Unit
Length	meter (m)
Mass	kilogram (kg)
Time	seconds (s)
Temperature	Kelvin (K)
Electric current	Ampere (A)
Amount of light	candela (cd)
Amount of matter	mole (mol)

Base SI units

Derived SI units

Force	Newton (N)
Energy (Work, Heat ..)	Joule (Nm)
Pressure	Pascale (Pa)
Temperature	Celsius (°C)

Any equation: ensure homogeneity of terms

$$P_{tot} = \frac{10 \text{ N}}{2 \text{ m}^2} + \frac{30 \text{ lbf}}{3 \text{ ft}^2}$$

Phases of matter

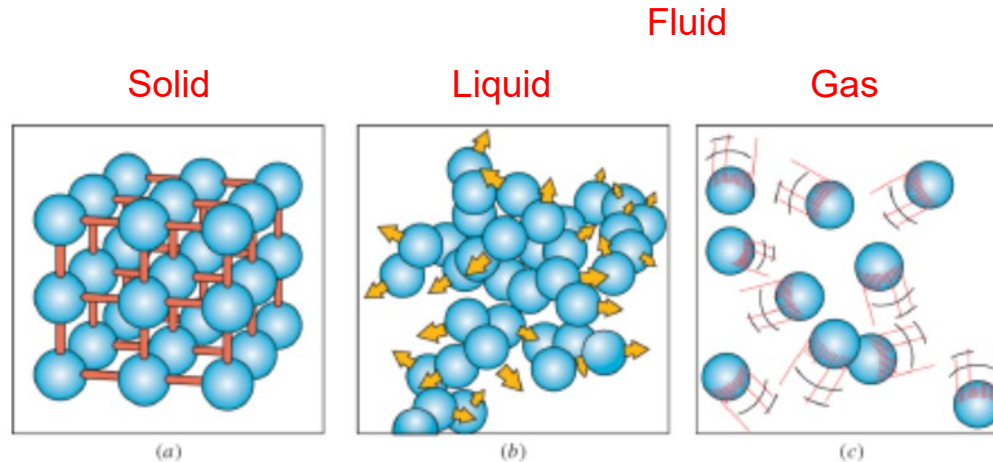


FIGURE 3–4

Melting
Freezing

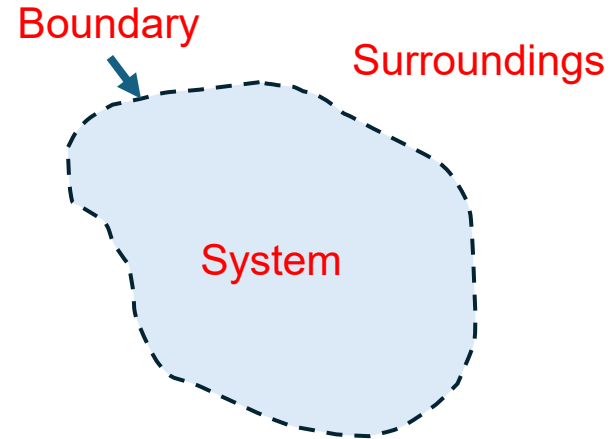
Vaporizing
Condensation

Sublimation
Deposition (resublimation)

Thermodynamic systems

Definition: A region of space containing matter in the form of molecules at a given point in time for which a thermodynamic analysis is performed.

- Boundary: physical or imaginary
- Systems and boundaries can change overtime (Moving boundaries)



Thermodynamic systems: examples

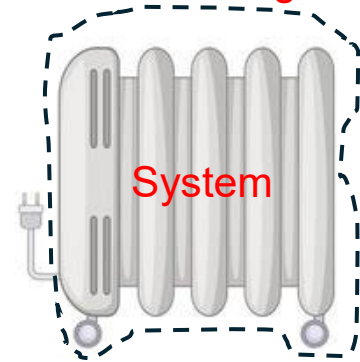
Surroundings



Surroundings



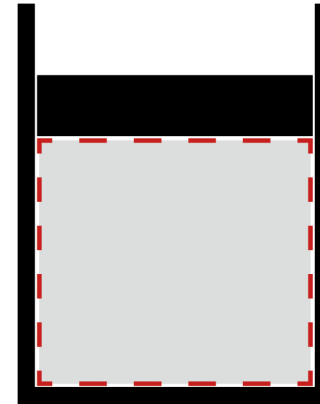
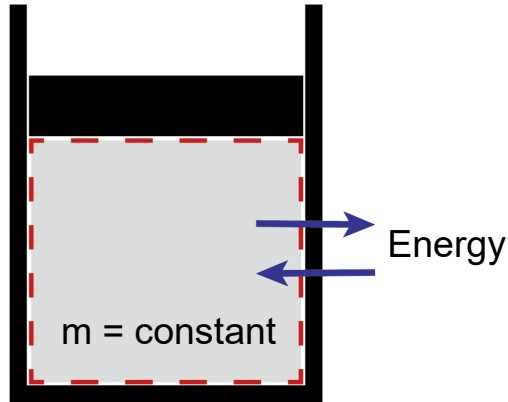
Surroundings



Before any thermodynamic analysis: define system boundary, choosing right system, can simplify analysis!

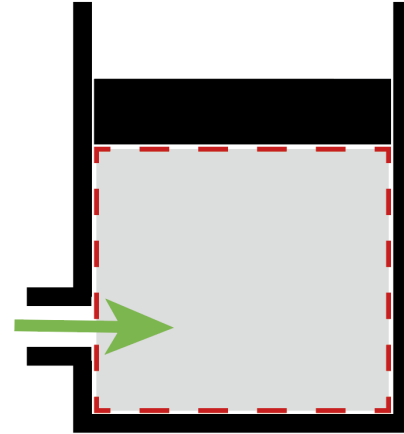
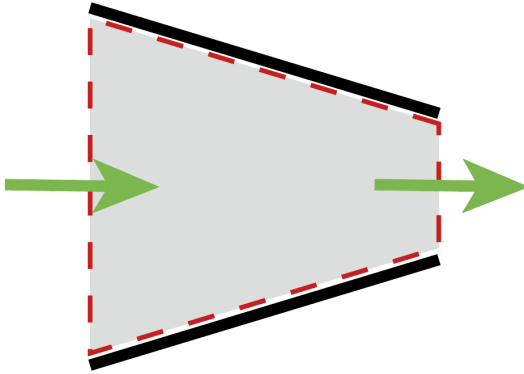
Thermodynamic systems

- **Closed system:** no mass flow across boundary.
- **Isolated system:** no energy flows across boundary



Thermodynamic systems

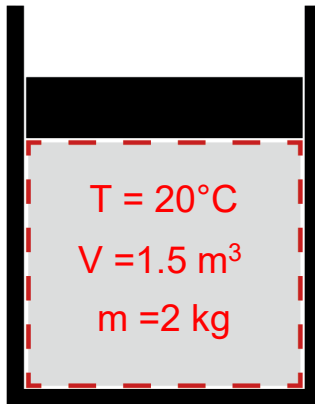
- Open system / Control volume



• Can also change over time, and show interactions W , Q etc.

Thermodynamic state and state properties

- Definition: state of the system that is not undergoing any change, which is fully defined by thermodynamic properties.

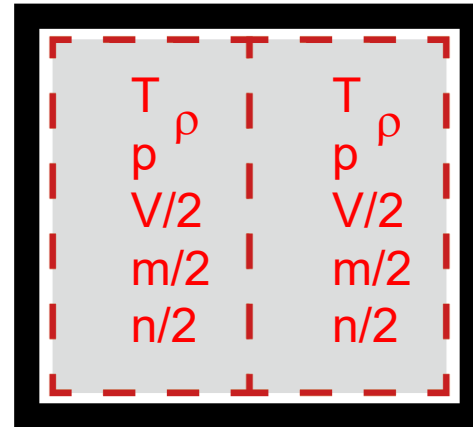
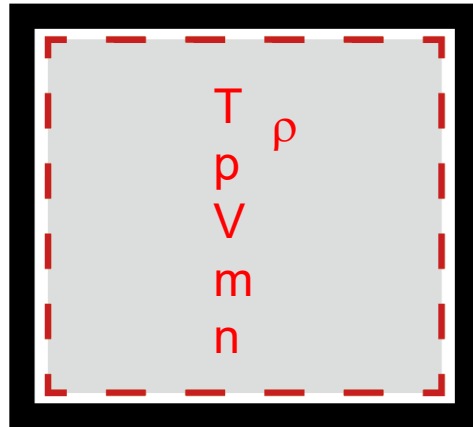


State properties

Pressure	p	[Pa]
Temperature	T	[K]
Volume	V	[m ³]
Mass	m	[kg]
Amount	n	[mol]

Thermodynamic state properties

- Intensive vs extensive and specific properties



Intensive

Extensive

Specific properties

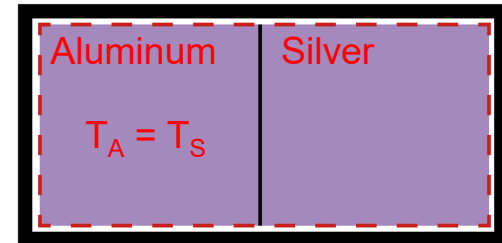
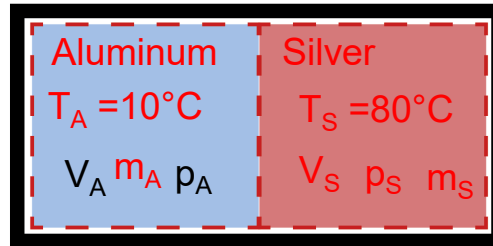
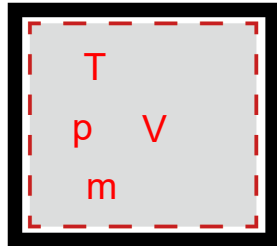
Specific volume

$$v = V/m$$

$[m^3/kg]$

Thermodynamic equilibrium

- Definition: state of balance, no unbalanced potential or driving force in the system.
- System experiences no changes when isolated from surroundings.
- Types: thermal, mechanical, phase, chemical



Thermodynamic state postulate

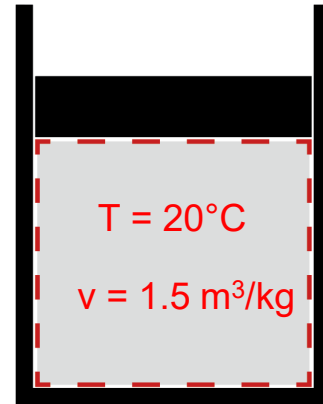
- Definition: A simple compressible system is completely defined by two independent intensive state properties.

$$T = f_1(p, v)$$

$$v = f_2(p, T)$$

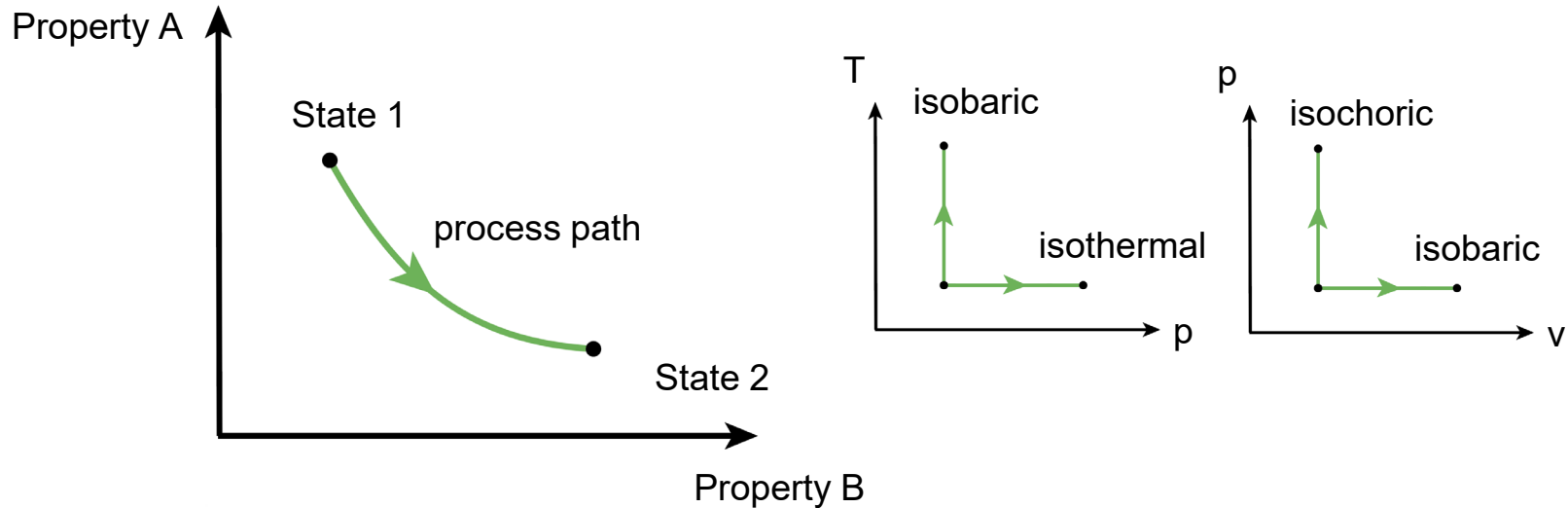
$$p = f_3(v, T)$$

$$0 = f(T, p, v)$$



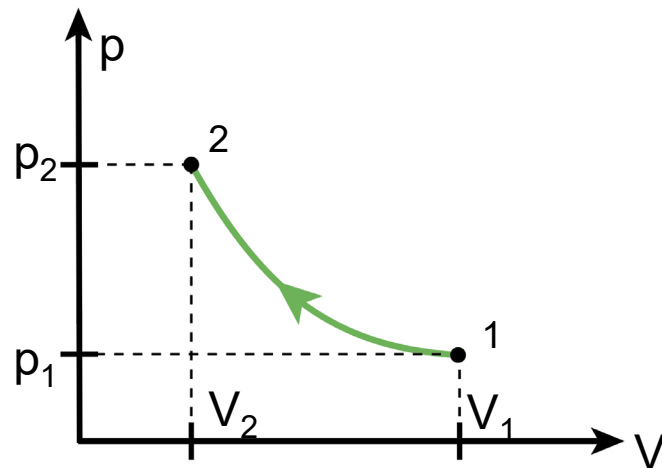
Thermodynamic process (I)

- Definition: Transformation from one state to another through interaction with the surroundings.



Thermodynamic process (II)

- Quasistatic process:
 - Process runs sufficiently slow so that systems passes through a series of equilibrium states.
- Non-quasistatic process:
 - Process runs too fast for equilibrium to be reached.



Summary

- Learned about thermodynamic systems, states, properties, equilibrium, processes, cycles.

Next week

- Density, pressure, temperature
- Different kinds of energy
- Learn about integral energy
- Energy transfer by work